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Risk models 1

Syllabus objectives

- 1.2 Compound distributions and their application in risk modelling
 - 1.2.1 Construct models appropriate for short-term insurance contracts in terms of the numbers of claims and the amounts of individual claims.
 - 1.2.2 Describe the major simplifying assumptions underlying the models in 1.2.1.
 - 1.2.3 Define a compound Poisson distribution and show that the sum of independent random variables each having a compound Poisson distribution also has a compound Poisson distribution.
 - 1.2.4 Derive the mean, variance and coefficient of skewness for compound binomial, compound Poisson and compound negative binomial random variables.

0 Introduction

In the first section of this chapter, we describe the main features of general insurance policies. There is no mathematics in this section, and you should be able to read it through fairly quickly in order to obtain a good overview of the different types of product available.

In the remaining sections of the chapter we introduce the idea of a *compound distribution*. We will define and use the compound Poisson, compound geometric, compound negative binomial and compound binomial distributions.

We will also start to look at two models, the *individual risk model* and the *collective risk model*, which are used to describe *aggregate claims*, ie the total claims that arise during a period from a group of policies.

In the simplest case of a life assurance benefit (often referred to as 'long-term business'), each policy can result in at most one claim and claims will be for amounts specified in advance (ie the sum assured). The benefit level may be the same for all policies or it may vary between policies.

In general insurance (often referred to as 'short-term business'), policies can give rise to more than one claim and the amounts will not usually be known in advance.

In this chapter we look at the theory of risk models. In the next chapter we explain how to adapt these models when reinsurance is in place.

1 General features of a product

1.1 Insurable interest

Generally, for a risk to be insurable:

- the policyholder must have an interest in the risk being insured, to distinguish between insurance and a wager
- a risk must be of a financial and reasonably quantifiable nature.

1.2 Insurable risk

Ideally risk events also need to meet the following criteria if they are to be insurable:

- **Individual risk events should be independent of each other.**
In practice we won't often get strict independence but a low correlation is desirable.
- **The probability of the event should be relatively small. In other words, an event that is nearly certain to occur is not conducive to insurance.**
For example, a house would not be insured if it stood on the edge of a crumbling cliff.
- **Large numbers of potentially similar risks should be pooled in order to reduce the variance and hence achieve more certainty.**
The similar risks should still be independent.
- **There should be an ultimate limit on the liability undertaken by the insurer.**
This would help the risk event meet the above criteria that it must be of a reasonably quantifiable nature.
- **Moral hazards should be eliminated as far as possible because these are difficult to quantify, result in selection against the insurer and lead to unfairness in treatment between one policyholder and another.**

Moral hazards occur when a person takes more risks because another party bears the cost of those risks.

However, the desire for income means that an insurer or reinsurer will usually be found to provide cover when these ideal criteria are not met.

Other characteristics that most general insurance products share are:

- **Cover is normally for a fixed period, most commonly one year, after which it has to be renegotiated. There is normally no obligation on the insurer or insured to continue the arrangement thereafter although in most cases a need for continuing cover may be assumed to exist.**
- **Claims are not of fixed amounts, and the amount of loss as well as the fact needs to be proved before a claim can be settled.**
- **A claim occurring does not bring the policy to an end.**

- **Claims may occur at any time during the policy period. Although there is normally a contractual obligation on the policyholder to report a claim to the insurer as quickly as possible, notification may take some time if the loss is not evident immediately. Settlement of the claim may take a long time if protracted legal proceedings are needed or if it is not straightforward to determine the extent of the loss. However, from the moment of the event giving rise to the claim the ultimate settlement amount is a liability of the insurer. Estimating the amounts of money that need to be reserved to settle these liabilities is one of the most important areas of actuarial involvement in general insurance.**

Classes of insurance in which claims tend to take a long time to settle are known as long-tail. Those which tend to take a short time to settle are known as short-tail, although the dividing line between the two categories is not always distinct.

2 Models for short-term insurance contracts

2.1 The basic model

Many forms of non-life insurance can be regarded as short-term contracts, for example motor insurance. Some forms of life insurance also fall into this category, for example group life and one-year term assurance policies.

A short-term insurance contract can be defined as having the following attributes:

- The policy lasts for a fixed, and relatively short, period of time, typically one year.
- The insurance company receives from the policyholder(s) a premium.
- In return, the insurer pays claims that arise during the term of the policy.

At the end of the policy's term the policyholder may or may not renew the policy. If it is renewed, the premium payable by the policyholder may or may not be the same as in the previous period.

The insurer may choose to pass part of the premium to a reinsurer. In return, the reinsurer will reimburse the insurer for part of the cost of the claims during the policy's term according to some agreed formula.

An important feature of a short-term insurance contract is that the premium is set at a level to cover claims arising during the (short) term of the policy only. This contrasts with life assurance policies where mortality rates increasing with age mean that the (level) annual premium in the early years would be more than sufficient to cover the expected claims in the early years. The excess amount would then be accumulated as a reserve to be used in the later years when the premium on its own would be insufficient to meet the expected cost of claims.

Now to be more specific, a short-term insurance contract covering a risk will be considered. A risk includes either a single policy or a specified group of policies. For ease of terminology the term of the contract is assumed to be one year, but it could equally well be any other short period, for example six months. The random variable S denotes the aggregate claims paid by the insurer in the year in respect of this risk. Models will be constructed for this random variable S . In Section 3 collective risk models will be studied. Later, in the next chapter, the idea of a collective risk model is extended to an individual risk model.

We will see shortly what these terms mean.

A first step in the construction of a collective risk model is to write S in terms of the number of claims arising in the year, denoted by the random variable N , and the amount of each individual claim. Let the random variable X_i denote the amount of the i th claim. Then:

$$S = \sum_{i=1}^N X_i \quad (19.1)$$

where the summation is taken to be zero if N is zero.

This decomposition of S allows consideration of claim numbers and claim amounts separately. A practical advantage of this is that the factors affecting claim numbers and claim amounts may well be different. Take motor insurance as an example. A prolonged spell of bad weather may have a significant effect on claim numbers but little or no effect on the distribution of individual claim amounts. On the other hand, inflation may have a significant effect on the cost of repairing cars, and hence on the distribution of individual claim amounts, but little or no effect on claim numbers.

This approach is referred to as a *collective risk model* because it is considering the *claims* arising from a group of policies taken as a whole, rather than by considering the claims arising from each individual policy.

The random variable S is the sum of a random number of random quantities, and is said to have a *compound distribution*.

Because compound distributions arise commonly in general insurance examples, the random variable N is often referred to as the 'number of claims' and the distribution of the random variables $X_1, X_2, \dots$ is referred to as the 'individual claim size distribution', even where the compound distribution arises in another context.

To define a compound distribution, we need to know:

- the distribution of N (which is a *discrete* distribution) and
- the distribution of the X_i 's (which may be *any* distribution).

If the distribution of the X_i 's is continuous, then S will have a *mixed distribution*, ie partly discrete and partly continuous. This is because of the possibility that $N = 0$.

The problems that will be studied are the derivation of the moments and distribution of S in terms of the moments and distributions of N and the X_i 's. Both will be studied with and without simple forms of reinsurance. The corresponding problems for the reinsurer will also be studied, ie the derivation of the moments and distribution of the aggregate claims paid in the year in respect of this risk by the reinsurer.

2.2 Discussion of the simplifications in the basic model

The model for short-term insurance described in the previous subsection contains a number of simplifications as compared to a real insurance operation. The first of these is that it is usually assumed that the moments, and sometimes the distributions, of N and the X_i 's are known with certainty. In practice these would probably be estimated from some relevant data.

For example, we might assume that claim amounts have a *Gamma*(500,4) distribution.

In practice it may not be possible to make such simple assumptions. For example:

- There may not be an appropriate theoretical distribution that models the distribution of claim amounts actually paid sufficiently well.
- Even if the shape of the distribution is satisfactory, appropriate parameter values may change over time, even in the short term.

- There may not be sufficient homogeneity in the portfolio. For example, different policies may produce claim amounts that have different sizes. This leads to the idea of a mixture distribution.

Another simplification is to assume, at least implicitly, that claims are settled more or less as soon as the incident causing the claim occurs, so that, for example, the insurer's profit is known at the end of the year. In practice, there will be at least a short delay in the settlement of claims and in some cases the delay can amount to many years. This will be especially true when the extent of the loss is difficult to determine, for example if it is to be decided in a court of law.

Delays will often lead to higher payments being made, owing to inflationary factors. (The relevant inflation rate may well be very different from that normally used to measure inflation.)

The model does not in general include any mention of expenses. The premium is assumed to pay the claims and include a loading for profit. In practice, the premium paid by the policyholder(s) will also include a loading for expenses. It is possible to include expenses in the model in a very simple way.

The simplest way to allow for expenses would be to use a claim size distribution that was artificially inflated to allow for some sort of claim expense amount (*eg* adding an extra 20%), although this might not give the right 'shape'. Alternatively we might express the random variable X as the sum of two other random variables, one to represent the actual claim amount and the other to represent the corresponding claim expense.

An important element in models for long-term insurance is a rate of interest since, as explained above, excess premium income would be invested to build up reserves. Interest is a relatively less important, but still important, feature of short-term insurance. It is possible to include interest in models for short-term insurance but it is more usual to ignore it, at least in elementary models.

We will ignore interest in the models used in this chapter.

There are a number of additional elements included when setting the premium to be charged to policyholders, including the policyholders' previous claims record and these are covered in Subject CP1 – Actuarial Practice. The allowance for policyholders' claim experience could be based on claim frequency or total claim amounts. This is beyond the scope of this subject.

In fact, we have already looked at models that make allowance for policyholders' claims experience in [Chapter 2](#).

2.3 Notation and assumptions

Throughout this chapter the following two important assumptions will be made:

- the random variables $\{X_i\}_{i=1}^N$ are independent and identically distributed
- the random variable N is independent of $\{X_i\}_{i=1}^N$.

In words these assumptions mean that:

1. the number of claims is not affected by the amount of individual claims
2. the amount of a given individual claim is not affected by the amount of any other individual claim
3. the distribution of the amounts of individual claims does not change over the (short) term of the policy.

Point 1 follows from the second of the two assumptions above. Points 2 and 3 follow from the first.

Throughout this chapter it will be assumed that all claims are for non-negative amounts, so that $P(X_i \leq x) = 0$ for $x < 0$. Many of the formulae in this chapter will be derived using the moment generating functions (from now on abbreviated to MGFs) of S , N and X_i . These MGFs will be denoted $M_S(t)$, $M_N(t)$ and $M_X(t)$, respectively, and will be assumed to exist for some positive values of the dummy variable t . The existence of the MGF of a non-negative random variable for positive values of t cannot generally be taken for granted; for example the MGFs of the Pareto and of the lognormal distributions do not exist for any positive value of t . However, all the formulae derived in this chapter with the help of MGFs can be derived, although less easily, without assuming the MGFs exist for positive values of t .

One method would be to use characteristic functions, $E(e^{itX})$, which don't have the same convergence problems as MGFs. However the Core Reading does not cover these.

$G(x)$ and $F(x)$ shall denote the distribution functions of S and X_i , respectively, so that:

$$G(x) = P(S \leq x) \text{ and } F(x) = P(X_i \leq x)$$

For convenience it will often be assumed that the density of $F(x)$ exists and it will be denoted $f(x)$. In cases where this density does not exist, so that X_i has a discrete or a mixed continuous/discrete distribution, expressions such as:

$$\int_0^\infty x f(x) dx$$

should be interpreted appropriately. The meaning should always be clear from the context.

The k th moment, ($k = 1, 2, 3, \dots$) of X_i about zero will be denoted m_k .

Using this notation:

$$E(X_i) = m_1 \quad \text{and} \quad \text{var}(X_i) = m_2 - m_1^2$$

3 The collective risk model

3.1 The collective risk model

Recall from Section 2.1 that S is represented as the sum of N random variables X_i , where X_i denotes the amount of the i th claim. Thus:

$$S = X_1 + X_2 + \cdots + X_N$$

and $S = 0$ if $N = 0$.

S is said to have a *compound distribution*.

Note that it is the number of claims, N , from the risk as a collective (as opposed to counting the number of claims from individual policies) that is being considered and this gives the name 'collective risk model'. Within this framework, expressions in general terms for the distribution function, mean, variance and MGF of S can be developed.

3.2 Distribution functions and convolutions

An expression for $G(x)$, the distribution function of S , can be derived by considering the event $\{S \leq x\}$. Note that if this event occurs, then one, and only one, of the following events must occur:

$$\{S \leq x \text{ and } N = 0\} \quad (\text{ie no claims})$$

$$\text{or} \quad \{S \leq x \text{ and } N = 1\} \quad (\text{ie one claim of amount } \leq x)$$

$$\text{or} \quad \{S \leq x \text{ and } N = 2\} \quad (\text{ie two claims which total } \leq x)$$

$$\vdots$$

$$\text{or} \quad \{S \leq x \text{ and } N = r\} \quad (\text{ie } r \text{ claims which total } \leq x)$$

$$\vdots$$

and so on. These events are mutually exclusive and exhaustive.

Thus:

$$\{S \leq x\} = \bigcup_{n=0}^{\infty} \{S \leq x \text{ and } N = n\}$$

and hence:

$$\begin{aligned} P(S \leq x) &= \sum_{n=0}^{\infty} P(S \leq x \text{ and } N = n) \\ &= \sum_{n=0}^{\infty} P(N = n) P(S \leq x \mid N = n) \end{aligned}$$



Question

A group of policies can give rise to at most two claims in a year. The probability function for the number of claims is as follows:

Number of claims, n	0	1	2
$P(N = n)$	0.6	0.3	0.1

Each claim is either for an amount of 1 or an amount of 2, with equal probability. Claim amounts are independent of one another and are independent of the number of claims.

Determine the distribution function of the aggregate annual claim amount, S .

Solution

S can take the values 0, 1, 2, 3 or 4.

S will only equal 0 if $N = 0$ and this has probability 0.6. So:

$$P(S \leq 0) = P(S = 0) = P(N = 0) = 0.6$$

S will equal 1 if there is one claim for amount 1. So:

$$\begin{aligned}
 P(S = 1) &= P(N = 1, X = 1) \\
 &= P(N = 1)P(X = 1) \quad \text{by independence} \\
 &= 0.3 \times 0.5 \\
 &= 0.15
 \end{aligned}$$

and:

$$P(S \leq 1) = P(S = 0) + P(S = 1) = 0.6 + 0.15 = 0.75$$

The other values of the CDF can be calculated in a similar way and are given below:

$$P(S \leq 2) = P(S \leq 1) + P(S = 2) = 0.75 + 0.3 \times 0.5 + 0.1 \times 0.5^2 = 0.925$$

$$P(S \leq 3) = P(S \leq 2) + P(S = 3) = 0.925 + 0.1 \times 2 \times 0.5^2 = 0.975$$

$$P(S \leq 4) = P(S \leq 3) + P(S = 4) = 1$$

The distribution of a sum of independent random variables can be found using *convolutions*.

If $Z = X + Y$, where X and Y are independent random variables with PDFs (or PFs) $f_X(x)$ and $f_Y(y)$, then $f_Z(z)$, the PDF (or PF) of Z , is called the *convolution* of X and Y .

This is written mathematically as $f_Z = f_X * f_Y$.

A formula for a convolution can be found by summing over all possible values of x and y that give a particular value z .

Finding a convolution

$$f_Z(z) = \sum_x f_X(x) f_Y(z-x) \quad \text{for discrete random variables}$$

$$f_Z(z) = \int f_X(x) f_Y(z-x) dx \quad \text{for continuous random variables}$$

'Sum or integrate over all values of x that could lead to a total of z .'

Similar formulae can be used to find the distribution function of a sum.

$$F_Z(z) = \sum_x f_X(x) F_Y(z-x) \quad \text{or} \quad \sum_x F_X(x) f_Y(z-x)$$

$$F_Z(z) = \int f_X(x) F_Y(z-x) dx \quad \text{or} \quad F_Z(z) = \int F_X(x) f_Y(z-x) dx$$



Question

Suppose that $N \sim \text{Poisson}(\lambda)$, $M \sim \text{Poisson}(\mu)$, and N and M are independent.

Use a convolution approach to derive the probability function of $N + M$.

Solution

Let $V = N + M$. Then, for $v = 0, 1, 2, \dots$:

$$\begin{aligned} P(V=v) &= \sum_{n=0}^v P(N=n)P(M=v-n) \\ &= \sum_{n=0}^v \frac{\lambda^n e^{-\lambda}}{n!} \frac{\mu^{v-n} e^{-\mu}}{(v-n)!} \\ &= \frac{e^{-(\lambda+\mu)}}{v!} \sum_{n=0}^v \frac{v!}{n!(v-n)!} \lambda^n \mu^{v-n} \\ &= \frac{e^{-(\lambda+\mu)}}{v!} \sum_{n=0}^v \binom{v}{n} \lambda^n \mu^{v-n} \\ &= \frac{e^{-(\lambda+\mu)}}{v!} (\lambda + \mu)^v \end{aligned}$$

This is the probability function of the $\text{Poisson}(\lambda + \mu)$ distribution. So $N + M \sim \text{Poisson}(\lambda + \mu)$.

This result can also be proved using MGFs.

We can now consider the distribution function for a compound distribution.

For convolutions of distribution functions, suppose that $\{X_i\}_{i=1}^n$ are independent and identically distributed (IID) random variables with common distribution function $F(x)$.

Then the distribution function of $\sum_{i=1}^n X_i$ is denoted by $F^{n*}(x)$, so that:

$$F^{n*}(x) = P(X_1 + X_2 + \dots + X_n \leq x)$$

Now note that if $N = n$, then S is the sum of a fixed number n , of random variables, $\{X_i\}_{i=1}^n$, and hence:

$$P(S \leq x \mid N = n) = F^{n*}(x)$$

where $F^{n*}(x)$ is the n -fold convolution of the distribution $F(x)$.

In other words, $F^{3*}(x)$ would be the convolution $F(x) * F(x) * F(x)$ etc.

(Note that $F^{1*}(x)$ is just $F(x)$ and, for convenience, $F^{0*}(x)$ is defined to equal 1 for all non-negative values of x . Otherwise $F^{0*}(x) = 0$.) Thus:

$$G(x) = P(S \leq x) = \sum_{n=0}^{\infty} P(N = n) F^{n*}(x) \quad (19.2)$$

Formula (19.2) is a general expression for the distribution function of S . Neither the distribution of N nor of X_i has been specified.

Note that when X_i is distributed on the positive integers it is easy to calculate $P(S = x)$ for $x = 1, 2, 3, \dots$ since:

$$\begin{aligned} P(S = x) &= G(x) - G(x-1) \\ &= \sum_{n=1}^{\infty} P(N = n) [F^{n*}(x) - F^{n*}(x-1)] \end{aligned}$$

$$\text{ie } P(S = x) = \sum_{n=1}^{\infty} P(N = n) f_x^{n*} \quad (19.3)$$

where $f_x^{n*} = F^{n*}(x) - F^{n*}(x-1)$ is the probability function of $\sum_{i=1}^n X_i$.

This is just saying that $f_x^{n*} = P(\sum_{i=1}^n X_i = x)$.

As in the case when X_i is a continuous random variable, $P(S = 0) = P(N = 0)$.

When the number of claims is large, and the claim amount distribution is not too skewed, we can approximate the distribution of S by a normal distribution with mean $E(S)$ and variance $\text{var}(S)$. We explain how to calculate moments of S in the next section.

3.3 Moments of compound distributions

To calculate the moments of S , conditional expectation results are used, conditioning on the number of claims, N . To find $E[S]$, apply the identity:

$$E[S] = E[E[S | N]]$$

Here we are using the conditional expectation formula, which is given on page 16 of the *Tables*.

Now $E[S | N = n] = \sum_{i=1}^n E[X_i] = nm_1$. Hence:

$$E[S | N] = Nm_1$$

and:

$$E[S] = E[Nm_1] = E[N]m_1 \quad (19.4)$$

This can also be written as follows:

$$E(S) = E(N)E(X)$$

where $E(X) = E(X_i)$, $i = 1, 2, \dots, N$.

Formula (19.4) has a very natural interpretation. It says that the expected aggregate claim amount is the product of the expected number of claims and the expected individual claim amount.

This formula is also given on page 16 of the *Tables*.



Question

If X has a Pareto distribution with parameters $\lambda = 400$ and $\alpha = 3$, and N has a *Poisson*(50) distribution, calculate the expected value of S .

Solution

The expected value is:

$$E(S) = E(N)E(X) = 50 \times \frac{400}{3-1} = 10,000$$

Variance

To find an expression for $\text{var}[S]$, apply the identity:

$$\text{var}[S] = E[\text{var}[S | N]] + \text{var}[E[S | N]]$$

Here we are using the conditional variance formula, which is given on page 16 of the *Tables*.

Since $E(S | N) = Nm_1$, we have:

$$\text{var}[S] = E[\text{var}[S | N]] + \text{var}[Nm_1]$$

$\text{var}[S | N]$ can be found by using the fact that individual claim amounts are independent.

Now:

$$\text{var}[S | N = n] = \text{var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{var}[X_i] = n(m_2 - m_1^2)$$

and so $\text{var}[S | N] = N(m_2 - m_1^2)$. Hence:

$$\text{var}[S] = E[N(m_2 - m_1^2)] + \text{var}[Nm_1]$$

$$\text{ie: } \text{var}[S] = E[N](m_2 - m_1^2) + \text{var}[N]m_1^2 \quad (19.5)$$

Alternatively, writing this solely in terms of means and variances:

$$\text{var}(S) = E(N)\text{var}(X) + \text{var}(N)[E(X)]^2$$

where $\text{var}(X) = \text{var}(X_i)$, $i = 1, 2, \dots, N$.

This formula can also be found on page 16 of the *Tables*. We will use it to determine the variances of the various compound distributions in the next few sections.

Unlike the expression for $E[S]$, formula (19.5) does not have a natural interpretation. The variance of S is expressed in terms of the mean and variance of both N and X_i .

However, this formula shows that the variability of the overall aggregate claim distribution is a function of both the variability in the number of claims and the variability in the claim amounts.

Moment generating function

The MGF of S is also found using conditional expectation. By definition, $M_S(t) = E[\exp(tS)]$, so:

$$M_S(t) = E[E[\exp(tS) | N]] \quad (19.6)$$

Again, we are conditioning on the number of claims, exactly as we did before.

Now $E[\exp(tS) | N = n] = E[\exp(tX_1 + tX_2 + \dots + tX_n)]$, and as $\{X_i\}_{i=1}^n$ are independent random variables:

$$E[\exp(tX_1 + tX_2 + \dots + tX_n)] = \prod_{i=1}^n E[\exp(tX_i)]$$

Also, since $\{X_i\}_{i=1}^n$ are identically distributed, they have common MGF, $M_X(t)$, so that:

$$\prod_{i=1}^n E[\exp(tX_i)] = \prod_{i=1}^n M_X(t) = [M_X(t)]^n$$

Hence:

$$E[\exp(tS) | N] = [M_X(t)]^N \quad (19.7)$$

These conditional expectations are random variables because they are functions of N .

Hence, inserting (19.7) in (19.6):

$$M_S(t) = E[M_X(t)^N] = E[\exp(N \log M_X(t))] = M_N(\log M_X(t)) \quad (19.8)$$

We can see this last step by observing that $E[\exp(N \log M_X(t))]$ is of the same form as $E(e^{Nt})$ but with t replaced by $\log M_X(t)$. So it is the MGF of N evaluated at $\log M_X(t)$.

Again, this is given on page 16 of the *Tables*.

Thus, the MGF of S is expressed in terms of the MGFs of N and of X_i . As with the previous results, the distributions of neither N nor of X_i have been specified.

A summary of the results for the mean, variance and MGF of S is given below.



Mean, variance and MGF of S

$$E(S) = E(N)E(X)$$

$$\text{var}(S) = E(N)\text{var}(X) + \text{var}(N)[E(X)]^2$$

$$M_S(t) = M_N(\log M_X(t))$$

There is one special case that is of some interest. This is when all claims are for the same fixed amount.

Example

Consider a portfolio of one-year term assurances each with the same sum assured. Suppose that the amount of a claim is B with probability one (assuming that a claim occurs at all), ie $P(X_i = B) = 1$ so that $m_1 = B$ and $m_2 = B^2$.

B is a constant here, not a random variable. So the expected value of an individual claim amount is B and its variance is 0.

Then S is distributed on $0, B, 2B, \dots$. In fact, $S = BN$ so:

$$P(S \leq Bx) = P(N \leq x)$$

Formulae (19.4) and (19.5) give the mean and variance of S , but as $S = BN$ it is easier to note that $E[S] = E[N]B$ and $\text{var}[S] = \text{var}[N]B^2$.

The next three sections consider compound distributions using various models for the number of claims, N .

3.4 The compound Poisson distribution

First consider aggregate claims when N has a Poisson distribution with mean λ denoted $N \sim \text{Poi}(\lambda)$. S then has a compound Poisson distribution with parameter λ , and $F(x)$ is the CDF of the individual claim amount random variable.

S is sometimes referred to as a compound Poisson random variable.

The results required for this distribution for N are:

$$E[N] = \text{var}[N] = \lambda$$

$$M_N(t) = \exp[\lambda(e^t - 1)]$$

Note that these results are given in the *Tables*.

These results can be combined with those of Section 3.1 as follows.

From (19.4):

$$E(S) = E(N)E(X) = \lambda E(X)$$

ie:

$$E[S] = \lambda m_1 \tag{19.9}$$

From (19.5):

$$\text{var}(S) = E(N)\text{var}(X) + \text{var}(N)[E(X)]^2 = \lambda \text{var}(X) + \lambda[E(X)]^2 = \lambda E(X^2)$$

ie:

$$\text{var}[S] = \lambda m_2 \tag{19.10}$$

and from (19.8):

$$M_S(t) = \exp[\lambda(M_X(t) - 1)] \tag{19.11}$$

The results for the mean and variance have a very simple form. Note that the variance of S is expressed in terms of the second moment of X_i about zero (and not in terms of the variance of X_i).

Note also that the formula for the skewness of S has a simple form when S is a compound Poisson random variable:

$$\text{skew}[S] = \lambda m_3 \quad (19.12)$$

ie:

$$\text{skew}(S) = \lambda E(X^3)$$

The easiest way to show that the third central moment of S is λm_3 is to use the cumulant generating function:

$$C_S(t) = \log M_S(t)$$

To determine the skewness, we differentiate it three times with respect to t and set $t = 0$, ie:

$$E[(S - \lambda m_1)^3] = \left. \frac{d^3}{dt^3} \log M_S(t) \right|_{t=0}$$

In other words:

$$\text{skew}(S) = C_S'''(0)$$

Recall also that:

$$E(S) = C_S'(0)$$

$$\text{var}(S) = C_S''(0)$$

Since $M_S(t) = \exp[\lambda(M_X(t) - 1)]$, it follows that:

$$\log M_S(t) = \lambda(M_X(t) - 1)$$

So:

$$\frac{d^3}{dt^3} \log M_S(t) = \lambda \left[\frac{d^3}{dt^3} M_X(t) - 1 \right]_{t=0} = \lambda m_3$$

ie $E[(S - \lambda m_1)^3] = \lambda m_3$

This is because $M_X'''(0) = E(X^3)$ for any random variable.

The coefficient of skewness of S is given by:

$$\frac{\text{skew}(S)}{[\text{var}(S)]^{3/2}}$$

Hence the coefficient of skewness = $\lambda m_3 / (\lambda m_2)^{3/2}$.

This result shows that the distribution of S is positively skewed, since m_3 is the third moment about zero of X_i and hence is greater than zero because X_i is a non-negative valued random variable. Note that the distribution of S is positively skewed even if the distribution of X_i is negatively skewed. The coefficient of skewness of S is

$\lambda m_3 / (\lambda m_2)^{3/2}$, and hence goes to 0 as $\lambda \rightarrow \infty$. Thus for large values of λ , the distribution of S is almost symmetric.



Mean, variance and skewness of a compound Poisson random variable

If $N \sim \text{Poisson}(\lambda)$, then S is a compound Poisson random variable and:

$$E(S) = \lambda E(X) = \lambda m_1$$

$$\text{var}(S) = \lambda E(X^2) = \lambda m_2$$

$$\text{skew}(S) = \lambda E(X^3) = \lambda m_3$$

These results are all given on page 16 of the *Tables*.

Sums of independent compound Poisson random variables

A very useful property of the compound Poisson distribution is that the sum of independent compound Poisson random variables is itself a compound Poisson random variable. A formal statement of this property is as follows.

Let $S_1, S_2, \dots, S_n$ be independent random variables. Suppose that each S_i has a compound Poisson distribution with parameter λ_i , and that the CDF of the individual claim amount random variable for each S_i is $F_i(x)$.

Define $A = S_1 + S_2 + \dots + S_n$. Then A has a compound Poisson distribution with parameter Λ , and $F(x)$ is the CDF of the individual claim amount random variable for A , where:

$$\Lambda = \sum_{i=1}^n \lambda_i \quad \text{and} \quad F(x) = \frac{1}{\Lambda} \sum_{i=1}^n \lambda_i F_i(x)$$

Recall that Λ is the capital form of the Greek letter λ .

This is a very important result.

To prove the result, first note that $F(x)$ is a weighted average of distribution functions and that these weights are all positive and sum to one. This means that $F(x)$ is a distribution function and this distribution has MGF:

$$M(t) = \int_0^{\infty} e^{tx} f(x) dx$$

where $f(x) = F'(x)$ is the PDF of the individual claim amount random variable for A .

So:

$$M(t) = \int_0^{\infty} \exp(tx) \frac{1}{\Lambda} \sum_{i=1}^n \lambda_i f_i(x) dx$$

where $f_i(x)$ is the density of $F_i(x)$. Hence:

$$M(t) = \frac{1}{\Lambda} \sum_{i=1}^n \lambda_i \int_0^{\infty} \exp\{tx\} f_i(x) dx = \frac{1}{\Lambda} \sum_{i=1}^n \lambda_i M_i(t) \quad (19.13)$$

where $M_i(t)$ is the MGF for the distribution with CDF $F_i(x)$.

Let $M_A(t)$ denote the MGF of A . Then:

$$M_A(t) = E[\exp(tA)] = E[\exp(tS_1 + tS_2 + \dots + tS_n)]$$

By independence of $\{S_i\}_{i=1}^n$:

$$M_A(t) = \prod_{i=1}^n E(\exp(tS_i))$$

As S_i is a compound Poisson random variable, its MGF is of the form given by formula (19.11), so:

$$E[\exp(tS_i)] = \exp[\lambda_i (M_i(t) - 1)]$$

Thus:

$$M_A(t) = \prod_{i=1}^n \exp\{\lambda_i (M_i(t) - 1)\} = \exp\left\{\sum_{i=1}^n \lambda_i (M_i(t) - 1)\right\}$$

ie:

$$M_A(t) = \exp\{\Lambda(M(t) - 1)\} \quad (19.14)$$

where:

$$\Lambda = \sum_{i=1}^n \lambda_i \quad \text{and} \quad M(t) = \frac{1}{\Lambda} \sum_{i=1}^n \lambda_i M_i(t)$$

By the one-to-one relationship between distributions and MGFs, formula (19.14) shows that A has a compound Poisson distribution with Poisson parameter Λ . By (19.13), the individual claim amount distribution has CDF $F(x)$.



Question

The distributions of aggregate claims from two risks, denoted by S_1 and S_2 , are as follows:

- S_1 has a compound Poisson distribution with parameter 100 and distribution function $F_1(x) = 1 - \exp(-x/\alpha)$, $x > 0$.
- S_2 has a compound Poisson distribution with parameter 200 and distribution function $F_2(x) = 1 - \exp(-x/\beta)$, $x > 0$.

Assuming that S_1 and S_2 are independent, determine the distribution of $S_1 + S_2$.

Solution

Let $S = S_1 + S_2$. Then S has a compound Poisson distribution with parameters $\Lambda = 300$ and $F(x)$, where:

$$F(x) = \frac{1}{3}F_1(x) + \frac{2}{3}F_2(x) = 1 - \frac{1}{3}\exp(-x/\alpha) - \frac{2}{3}\exp(-x/\beta)$$

We can use R to simulate values from a compound Poisson distribution.



The R code to simulate 10,000 values from a compound Poisson distribution with parameter 1,000 and a gamma claims distribution with $\alpha = 750$ and $\lambda = 0.25$ is:

```
set.seed(123)
n <- rpois(10000, 1000)
s <- numeric(10000)
for(i in 1:10000)
{ x <- rgamma(n[i], shape=750, rate=0.25)
  s[i] <- sum(x) }
```

We can obtain a mean of 2,997,651, a standard deviation of 93,719.71 and a coefficient of skewness of 0.02655921 as follows:

```
mean(s)
sd(s)
skewness <- sum((s - mean(s))^3) / length(s)
coeff.of.skew <- skewness / var(s)^(3/2)
```

We can estimate $P(S > 3,000,000)$ to be 0.4881 as follows:

```
length(s[s > 3000000]) / length(s)
```

Finally we could estimate the 90th percentile to be 3,115,719 as follows::

```
quantile(s, 0.9)
```

We can plot a histogram of the compound distribution using the `hist` function and an empirical density function using `density` in the `plot` function. We can then superimpose a normal or other distribution to see if they provide a good approximation.

However, a better way to check the fit with a normal distribution is to use the `qqnorm` function:

```
qqnorm(<simulated values>)
```

or the `qqplot` function to compare the sample data to simulated values from a fitted model distribution:

```
qqplot(<simulated theoretical values>,
       <simulated compound distribution values>)
```

Note we have used `set.seed(123)` so you can obtain the same values as this example.

3.5 The compound binomial distribution

Under certain circumstances, the binomial distribution is a natural choice for N . For example, under a group life insurance policy covering n lives, the distribution of the number of deaths in a year is binomial if it is assumed that each insured life is subject to the same mortality rate, and that lives are independent with respect to mortality.

The notation $N \sim \text{Bin}(n, p)$ is used to denote the binomial distribution for N . The key results for this distribution are:

$$E[N] = np$$

$$\text{var}[N] = np(1-p)$$

$$M_N(t) = (pe^t + 1 - p)^n$$

Note that these results are given in the *Tables*.

However, the notation for the MGF is slightly different.

When N has a binomial distribution, S has a compound binomial distribution. One important point about choosing the binomial distribution for N is that there is an upper limit, n , to the number of claims.

Expressions for the mean, variance and MGF of S are now found in terms of n , p , m_1 , m_2 and $M_X(t)$ when $N \sim \text{Bin}(n, p)$.

There is no need to memorise the formulae in this section. However, it is important to be able to derive them.

Formula (19.4) gives the mean:

$$E(S) = E(N)E(X)$$

$$\Rightarrow E[S] = npm_1 \quad (19.15)$$

Formula (19.5) gives the variance:

$$\begin{aligned}\text{var}(S) &= E(N) \text{var}(X) + \text{var}(N)[E(X)]^2 \\ \Rightarrow \text{var}[S] &= np(m_2 - m_1^2) + np(1-p)m_1^2 \\ &= npm_2 - np^2m_1^2\end{aligned}\tag{19.16}$$

Lastly, formula (19.8) gives the MGF:

$$\begin{aligned}M_S(t) &= M_N(\log M_X(t)) \\ \Rightarrow M_S(t) &= (pM_X(t) + 1 - p)^n\end{aligned}$$

We can also find expressions for the skewness and the coefficient of skewness.

The third central moment is found from the cumulant generating function:

$$C_S(t) = \ln M_S(t)$$

In the next few steps, liberal use is made of the chain rule $\left(\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}\right)$ and the product rule for differentiation $\left(\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}\right)$. The third derivative of the cumulant generating function is:

$$\begin{aligned}\frac{d^3}{dt^3} \log M_S(t) &= \frac{d^3}{dt^3} n \log(pM_X(t) + q) \quad \text{where } q = 1 - p \\ &= \frac{d^2}{dt^2} \left\{ np \left(\frac{d}{dt} M_X(t) \right) (pM_X(t) + q)^{-1} \right\} \\ &= \frac{d}{dt} \left\{ np \left(\frac{d^2}{dt^2} M_X(t) \right) (pM_X(t) + q)^{-1} - n \left(p \frac{d}{dt} M_X(t) \right)^2 (pM_X(t) + q)^{-2} \right\} \\ &= np \left(\frac{d^3}{dt^3} M_X(t) \right) (pM_X(t) + q)^{-1} \\ &\quad - 3np^2 \left(\frac{d^2}{dt^2} M_X(t) \right) (pM_X(t) + q)^{-2} \left(\frac{d}{dt} M_X(t) \right) \\ &\quad + 2n \left(p \frac{d}{dt} M_X(t) \right)^3 (pM_X(t) + q)^{-3}\end{aligned}$$

Setting $t = 0$ gives:

$$\text{skew}(S) = E[(S - npm_1)^3] = npm_3 - 3np^2m_2m_1 + 2np^3m_1^3\tag{19.17}$$

The coefficient of skewness is then given by:

$$\frac{\text{skew}(S)}{[\text{var}(S)]^{3/2}} = \frac{npm_3 - 3np^2m_2m_1 + 2np^3m_1^3}{(npm_2 - np^2m_1^2)^{3/2}}$$

It can be deduced from formula (5.17) that it is possible for the compound binomial distribution to be negatively skewed. The simplest illustration of this fact is when all claims are of (a fixed) amount B . Then $S = BN$ and:

$$E[(S - E[S])^3] = B^3 E[(N - E[N])^3]$$

ie:

$$\text{skew}(S) = B^3 \text{skew}(N)$$

So the coefficient of skewness of S is a multiple of that for N .

In fact:

$$\text{coeff of skew}(S) = \frac{\text{skew}(S)}{[\text{var}(S)]^{3/2}} = \frac{B^3 \text{skew}(N)}{[B^2 \text{var}(N)]^{3/2}} = \text{coeff of skew}(N)$$

If $p > 0.5$, then the binomial distribution for N is negatively skewed.

So the coefficient of skewness of S will also be negative in this case.



Question

Determine an expression for the MGF of the aggregate claim amount random variable if the number of claims has a $\text{Bin}(100, 0.01)$ distribution and individual claim sizes have a $\text{Gamma}(10, 0.2)$ distribution.

Solution

Since $N \sim \text{Bin}(100, 0.01)$ and $X \sim \text{Gamma}(10, 0.2)$, we have:

$$M_N(t) = (0.99 + 0.01e^t)^{100} \quad \text{and} \quad M_X(t) = \left(1 - \frac{t}{0.2}\right)^{-10} = (1 - 5t)^{-10}$$

So:

$$\begin{aligned} M_S(t) &= M_N[\log M_X(t)] \\ &= \left[0.99 + 0.01e^{\log M_X(t)}\right]^{100} \\ &= \left[0.99 + 0.01M_X(t)\right]^{100} \\ &= \left[0.99 + 0.01(1 - 5t)^{-10}\right]^{100} \end{aligned}$$

3.6 The compound negative binomial distribution

The final choice of distribution for N is the negative binomial distribution, which has probability function:

$$P(N = n) = \binom{k+n-1}{n} p^k q^n \quad \text{for } n = 0, 1, 2, \dots$$

This is the Type 2 negative binomial distribution. See page 9 of the *Tables*.

The Type 1 negative binomial distribution has probability function:

$$P(N = n) = \binom{n-1}{k-1} p^k q^{n-k} \quad \text{for } n = k, k+1, k+2, \dots$$

It is not likely to be appropriate here, unless there is a specific reason why the number of claims must be at least k .

The parameters of the distribution are k (> 0) and p , where $p + q = 1$ and $0 < p < 1$. This distribution is denoted by $NB(k, p)$. When $N \sim NB(k, p)$:

$$E[N] = \frac{kq}{p}$$

$$\text{var}[N] = \frac{kq}{p^2}$$

$$M_N(t) = p^k (1 - qe^t)^{-k}$$

The special case $k = 1$ leads to the geometric distribution. Once again, note that these results are given in the *Tables*.

The negative binomial distribution is an alternative to the Poisson distribution for N .

This is because the negative binomial distribution can take any non-negative integer value, unlike the binomial distribution which has an upper limit.

One advantage that the negative binomial distribution has over the Poisson distribution is that its variance exceeds its mean. These two quantities are equal for the Poisson distribution. Thus, the negative binomial distribution may give a better fit to a data set which has a sample variance in excess of the sample mean. This is often the case in practice. In the next chapter a situation leading to the negative binomial distribution for N is discussed. When N has a negative binomial distribution, S has a compound negative binomial distribution.

Expressions for the mean, variance and MGF of S when $N \sim NB(k, p)$ come immediately from formulae (19.4), (19.5) and (19.8):

$$E(S) = E(N)E(X) \Rightarrow E[S] = \frac{kq}{p} m_1$$

$$\text{var}(S) = E(N) \text{var}(X) + \text{var}(N)[E(X)]^2 \Rightarrow \text{var}[S] = \frac{kq}{p} (m_2 - m_1^2) + \frac{kq}{p^2} m_1^2$$

Multiplying out the brackets and regrouping the terms, we see that:

$$\begin{aligned} \frac{kq}{p} (m_2 - m_1^2) + \frac{kq}{p^2} m_1^2 &= \frac{kq}{p} m_2 - \frac{kq}{p} m_1^2 + \frac{kq}{p^2} m_1^2 \\ &= \frac{kq}{p} m_2 - \frac{kpq}{p^2} m_1^2 + \frac{kq}{p^2} m_1^2 \\ &= \frac{kq}{p} m_2 + \frac{kq - kpq}{p^2} m_1^2 \\ &= \frac{kq}{p} m_2 + \frac{kq(1-p)}{p^2} m_1^2 \\ &= \frac{kq}{p} m_2 + \frac{kq^2}{p^2} m_1^2 \end{aligned}$$

So:

$$\text{var}[S] = \frac{kq}{p} m_2 + \frac{kq^2}{p^2} m_1^2$$

and:

$$M_S(t) = M_N(\log M_X(t)) \Rightarrow M_S(t) = \frac{p^k}{(1 - qM_X(t))^k}$$

As before, the third central moment of S can be found from the cumulant generating function of S , as follows:

$$\begin{aligned} \frac{d}{dt} \log M_S(t) &= \frac{d}{dt} (k \log p - k \log [1 - qM_X(t)]) \\ &= \frac{kq}{1 - qM_X(t)} \left(\frac{d}{dt} M_X(t) \right) \end{aligned}$$

Then:

$$\frac{d^2}{dt^2} \log M_S(t) = kq^2 \left(\frac{d}{dt} M_X(t) \right)^2 \frac{1}{(1 - qM_X(t))^2} + \frac{kq}{1 - qM_X(t)} \left(\frac{d^2}{dt^2} M_X(t) \right)$$

and:

$$\begin{aligned} \frac{d^3}{dt^3} \log M_S(t) &= 3kq^2 \left(\frac{d}{dt} M_X(t) \right) \left(\frac{d^2}{dt^2} M_X(t) \right) \frac{1}{(1 - qM_X(t))^2} \\ &+ \frac{2kq^3}{(1 - qM_X(t))^3} \left(\frac{d}{dt} M_X(t) \right)^3 + \frac{kq}{1 - qM_X(t)} \left(\frac{d^3}{dt^3} M_X(t) \right) \end{aligned}$$

Setting $t = 0$ in the third derivative gives:

$$\text{skew}(S) = E[(S - E[S])^3] = \frac{3kq^2 m_1 m_2}{p^2} + \frac{2kq^3 m_1^3}{p^3} + \frac{kq m_3}{p} \quad (19.18)$$

The parameters k and p are positive, as are the moments of X . It therefore follows from formula (19.18) that the compound negative binomial distribution is positively skewed. The coefficient of skewness can be found from $E((S - E(S))^3) / (\text{var}(S))^{3/2}$.



Question

The distribution of the number of claims from a motor portfolio is negative binomial with parameters $k=4,000$ and $p=0.9$. The claim size distribution is Pareto with parameters $\alpha=5$ and $\lambda=1,200$. Calculate the mean and standard deviation of the aggregate claim distribution.

Solution

The first two moments of the Pareto distribution are:

$$\begin{aligned} m_1 &= E(X) = \frac{\lambda}{\alpha - 1} = \frac{1,200}{4} = 300 \\ m_2 &= E(X^2) = \frac{\alpha \lambda^2}{(\alpha - 1)^2 (\alpha - 2)} + m_1^2 = \frac{5 \times 1,200^2}{4^2 \times 3} + 300^2 = 240,000 \end{aligned}$$

So, using the formulae for the mean and variance of a compound negative binomial distribution:

$$\begin{aligned} E(S) &= \frac{kq}{p} \times m_1 = \frac{4,000 \times 0.1}{0.9} \times 300 = 133,333 \\ \text{var}(S) &= \frac{kq}{p} m_2 + \frac{kq^2}{p^2} m_1^2 = \frac{4,000 \times 0.1}{0.9} \times 240,000 + \frac{4,000 \times 0.1^2}{0.9^2} \times 300^2 = 111,111,111 \end{aligned}$$

So the standard deviation is 10,541.

4 Appendix

There is some repetition in the Core Reading in Section 3.3. To improve the flow of the chapter, we have removed the repeated section from the main part of the text and placed it below.

Let $S = X_1 + X_2 + \dots + X_N$ (and $S = 0$ if $N = 0$) where the X_i 's are independent, identically distributed (as a variable X) and are also independent of N . S is said to have a *compound distribution*.

Illustration: N is the number of claims which arise in a portfolio of business and X_i is the amount of the i th claim. S is the total claim amount.

The mean and variance of S are easily found:

$$E(S | N = n) = E(X_1 + \dots + X_N | N = n) = E(X_1 + \dots + X_n) = nE(X)$$

Similarly:

$$\text{var}(S | N = n) = n \text{var}(X)$$

Therefore:

$$E(S) = E[E(S | N)] = E[NE(X)] = E(N)E(X)$$

ie:

$$\mu_S = \mu_N \mu_X$$

and:

$$\begin{aligned} \text{var}(S) &= E[\text{var}(S | N)] + \text{var}[E(S | N)] \\ &= E[N \text{var}(X)] + \text{var}[NE(X)] \\ &= E(N) \text{var}(X) + \text{var}(N)[E(X)]^2 \end{aligned}$$

ie:

$$\sigma_S^2 = \mu_N \sigma_X^2 + \sigma_N^2 \mu_X^2$$

The MGF of S is given by:

$$M_S(t) = E(e^{tS}) = E[E(e^{tS} | N)]$$

and:

$$\begin{aligned} E(e^{tS} | N = n) &= E[\exp(t(X_1 + X_2 + \dots + X_N)) | N = n] \\ &= E[\exp(t(X_1 + X_2 + \dots + X_n))] \\ &= \prod E[\exp(tX_i)] = [M_X(t)]^n \end{aligned}$$

Therefore:

$$M_S(t) = E\left[(M_X(t))^N\right] = E\left[\exp(N \log M_X(t))\right] = M_N(\log M_X(t))$$

Compound Poisson distribution

An important illustration is provided by the compound Poisson distribution, which is the case in which $N \sim \text{Poisson}(\lambda)$. In this case $\mu_N = \sigma_N^2 = \lambda$.

Properties:

$$E(S) = \lambda E(X)$$

$$\text{var}(S) = \lambda \text{var}(X) + \lambda [E(X)]^2 = \lambda E(X^2)$$

$$M_N(t) = \exp\left[\lambda(e^t - 1)\right]$$

so:

$$M_S(t) = \exp\left[\lambda(M_X(t) - 1)\right]$$

from which the mean and variance can be obtained and the results above verified.

Chapter 19 Summary

Insurable risks

For a risk to be insurable the policyholder should have an interest in the risk being insured to distinguish between insurance and a wager, and it should be of a financial and reasonably quantifiable nature. Ideally, risk events should:

- be independent
- have low probability of occurring
- be pooled with similar risks
- have an ultimate liability
- avoid moral hazards.

Characteristics of general insurance products

Most general insurance contracts share the following characteristics:

- Cover is normally for a fixed period, typically a year, after which it needs to be renegotiated.
- There is usually no obligation to continue cover although in most cases a need for continuing cover may be assumed to exist.
- Claims are not of fixed amounts.
- The existence of a claim and its amount have to be proved before a claim can be settled.
- A claim occurring does not bring the policy to an end.
- Claims that take a long time to settle are known as long-tailed and those that take a short time to settle are known as short-tailed.

Features of short-term insurance contracts

A short-term insurance contract can be defined as having the following attributes:

- The policy lasts for a fixed, and relatively short time period, typically one year.
- The insurance company receives a premium from the policyholder.
- In return the insurer pays claims that arise during the term of the policy.
- At the end of the policy term, the policyholder may or may not renew the policy. If it is renewed, the premium may or may not be the same as in the previous period.
- The insurer may pass part of the premium to a reinsurer, who, in return, will reimburse the insurer for part of the claims cost.

Collective risk model

Aggregate claim amounts may be modelled using a compound distribution. The aggregate claim amount S is the sum of a random number of IID random variables:

$$S = X_1 + X_2 + \dots + X_N$$

where S is taken to be zero if $N = 0$. We assume that the random variable N is independent of the random variables X_i so that the distributions of the claim numbers and the individual claim amounts can be analysed separately. The distribution of S is said to be a compound distribution.

Other simplifying assumptions include:

- The moments (and sometimes the distributions) of N and X_i are known.
- Claims are settled more or less as soon as the claims occur.
- Expenses and investment returns are ignored.

Specific types of compound distributions include the compound Poisson, compound binomial, compound negative binomial, and compound geometric. Formulae for the MGF and the moments of a compound random variable are given on page 16 of the *Tables*.

Convolutions

If $Z = X + Y$, and X and Y are independent, then:

$$P(Z = z) = \sum_x P(X = x)P(Y = z - x) \text{ if } X \text{ and } Y \text{ are discrete}$$

$$f_Z(z) = f_X * f_Y(z) = \int_{-\infty}^{\infty} f_X(x)f_Y(z - x)dx \text{ if } X \text{ and } Y \text{ are continuous}$$

Sums of independent compound Poisson random variables

Let $S_1, S_2, \dots, S_n$ be a set of independent random variables where S_i has a compound Poisson distribution with parameter λ_i and $F_i(x)$ is the CDF of the individual claim amount random variable for S_i . Then $A = S_1 + \dots + S_n$ is compound Poisson with parameter $\Lambda = \sum \lambda_i$. The CDF of the individual claim amount random variable for A is:

$$F(x) = \frac{1}{\Lambda} \sum \lambda_i F_i(x)$$

The MGF of the individual claim amounts for A is:

$$M(t) = \frac{1}{\Lambda} \sum \lambda_i M_i(t)$$

where $M_i(t)$ is the MGF of the individual claim amounts for S_i .



Chapter 19 Practice Questions

- 19.1 (i) State the two conditions that must hold for a risk to be insurable.
- (ii) List five other risk criteria that would be considered desirable by a general insurer.
- 19.2 A group of policies can give rise to at most 2 claims. The probabilities of 0, 1 or 2 claims are $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$ respectively. Claim amounts are IID $U(0,10)$ random variables. Let S denote the aggregate claim amount random variable.

Sketch the frequency distribution of S .

- 19.3 The random variable S has a compound Poisson distribution with Poisson parameter 4. The individual claim amounts are either 1, with probability 0.3, or 3, with probability 0.7.

Calculate the probability that $S = 4$.

- 19.4 A compound random variable $S = X_1 + X_2 + \dots + X_N$ has claim number distribution:

$$P(N = n) = 9(n+1)4^{-n-2}, \quad n = 0, 1, 2, \dots$$

The individual claim size random variable, X , is exponentially distributed with mean 2.

Calculate $E(S)$ and $\text{var}(S)$.

- 19.5 Write down a formula for the MGF of a compound Poisson distribution with individual claim size distribution $\text{Gamma}(\alpha, \beta)$ and Poisson parameter λ .

- 19.6 S_1 and S_2 are independent random variables each with a compound Poisson distribution. The distribution of S_i , $i = 1, 2$, has Poisson parameter λ_i and individual claim amount distribution $F_i(x)$.

Which one of the following statements about the distribution of $S_1 + S_2$ is correct?

- A $S_1 + S_2$ has a compound Poisson distribution with Poisson parameter $\lambda_1 \lambda_2$ and individual claim amount distribution $F_1(x) + F_2(x)$.
- B $S_1 + S_2$ has a compound Poisson distribution with Poisson parameter $(\lambda_1 + \lambda_2)$ and individual claim amount distribution $(F_1(x) + F_2(x))/2$.
- C $S_1 + S_2$ has a compound Poisson distribution with Poisson parameter $(\lambda_1 + \lambda_2)$ and individual claim amount distribution $(\lambda_1 F_1(x) + \lambda_2 F_2(x))/(\lambda_1 + \lambda_2)$.
- D $S_1 + S_2$ does not have a compound Poisson distribution.

- 19.7** Claims on a group of policies of a certain type arise as a Poisson process with parameter λ_1 .
 Claims on a second, independent, group of policies arise as a Poisson process with parameter λ_2 .
 The aggregate claim amounts on the respective groups are denoted S_1 and S_2 .

Exam style

Using moment generating functions (or otherwise), show that S (the sum of S_1 and S_2) also has a compound Poisson distribution and hence derive the Poisson parameter for S . [4]

- 19.8** The aggregate claim amount from a portfolio has a compound negative binomial distribution.

Exam style

- (i) Show that if $S = X_1 + \dots + X_N$, then:

$$M_S(t) = M_N[\log M_X(t)] \quad [3]$$

- (ii) If N has Type 2 negative binomial distribution with $k = 2$ and $p = 0.9$, and X has a gamma distribution with $\alpha = 10$ and $\lambda = 0.1$, determine an expression for $M_S(t)$. [2]

- (iii) (a) Calculate the mean and variance of S .
 (b) Using a suitable approximation, estimate the aggregate amount which will be exceeded with probability 0.1%. [4]

- (iv) The insurer in fact has 100 identical independent portfolios of this type. Let:

$$T = S_1 + \dots + S_{100}$$

- (a) Determine the moment generating function for T .
 (b) Using a normal approximation, estimate the total aggregate claim amount from the whole business which will be exceeded with probability 0.1%.
 (c) Comment on your answers to parts (iii)(b) and (iv)(b). [4]

[Total 13]



Chapter 19 Solutions

19.1 (i) *Criteria for an insurable risk*

The two conditions are:

- The policyholder must have an interest in the risk being insured.
- The risk must be of a financial and reasonably quantifiable nature.

(ii) *Other desirable features of a risk*

Other desirable features are:

- Individual risks should be independent of one another.
- The probability that the insured event will occur should be small.
- Large numbers of similar risks should be pooled in order to reduce the variance and achieve greater certainty.
- The insurer's liability should be limited.
- Moral hazards should be eliminated as far as possible since these are difficult to quantify, result in selection against the insurer and lead to unfairness in the treatment of some policyholders.

19.2 If $N = 0$, ie there are no claims, then $S = 0$. So there is a point mass (or a 'blob' of probability) at $S = 0$.

If $N = 1$, ie there is exactly one claim, then S has a $U(0, 10)$ distribution. This will happen with probability $\frac{1}{4}$.

If $N = 2$, ie there are exactly 2 claims, then $S = X_1 + X_2$ where X_1 and X_2 are independent random variables with PDFs:

$$f_{X_1}(x) = f_{X_2}(x) = \begin{cases} 0.1 & \text{if } 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

The PDF of S is:

$$f_S(s) = \int_{-\infty}^{\infty} f_{X_1}(s-x)f_{X_2}(x)dx = 0.1 \int_0^{10} f_{X_1}(s-x)dx$$

The integrand is 0.1 if $0 \leq s-x \leq 10$ and 0 otherwise. So for $0 \leq s \leq 10$:

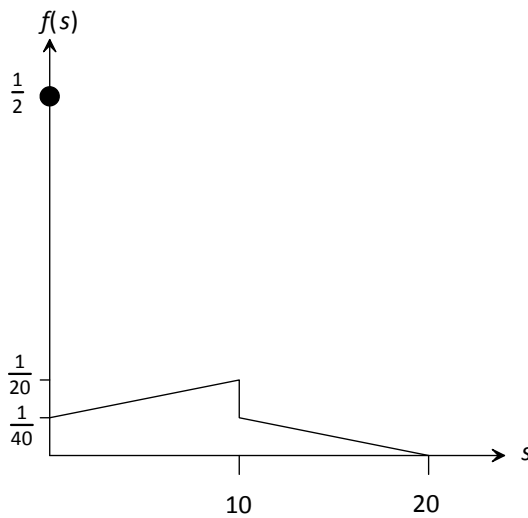
$$f_S(s) = 0.1 \int_0^s 0.1 dx = 0.01s$$

and for $10 \leq s \leq 20$:

$$f_S(s) = 0.1 \int_{s-10}^{10} 0.1 dx = 0.2 - 0.01s$$

So for $N = 2$, S has a symmetrical triangular shaped distribution on the interval $(0, 20)$.

A graph of the distribution is shown below:



This graph is the combination of a blob at zero, a uniform distribution on $(0,10)$ and a triangular distribution on $(0,20)$.

19.3 We need to consider how we could get an aggregate claim amount of 4. This could happen in two ways:

- (a) 2 claims, one for 1 and one for 3
- (b) 4 claims, all for an amount of 1.

The probability of this happening is:

$$P(S = 4) = P(N = 2)P(X_1 = 1)P(X_2 = 3) + P(N = 2)P(X_1 = 3)P(X_2 = 1) \\ + P(N = 4)P(X_1 = 1)P(X_2 = 1)P(X_3 = 1)P(X_4 = 1)$$

Since the X_i 's are identical, this simplifies to:

$$P(S = 4) = 2P(N = 2)P(X = 1)P(X = 3) + P(N = 4)[P(X = 1)]^4 \\ = 2 \times \frac{e^{-4} 4^2}{2!} \times 0.3 \times 0.7 + \frac{e^{-4} 4^4}{4!} \times 0.3^4 \\ = 0.06312$$

19.4 The probability function of N can be written as:

$$P(N = n) = 9(n+1)4^{-n-2} = \binom{n+1}{n} (3/4)^2 (1/4)^n$$

We can see from this formula that N has a Type 2 negative binomial distribution with parameters $k = 2$ and $p = 3/4$.

Hence:

$$E(N) = \frac{kq}{p} = \frac{2 \times 1/4}{3/4} = 2/3$$

and: $\text{var}(N) = \frac{kq}{p^2} = \frac{2 \times 1/4}{(3/4)^2} = 8/9$

The individual claim amounts have an exponential distribution with $\lambda = 1/2$. So the mean and variance of the individual claims are:

$$E(X) = \frac{1}{\lambda} = 2 \quad \text{and} \quad \text{var}(X) = \frac{1}{\lambda^2} = 4$$

Hence:

$$E(S) = E(N) E(X) = \frac{2}{3} \times 2 = \frac{4}{3}$$

and: $\text{var}(S) = E(N) \text{var}(X) + \text{var}(N)[E(X)]^2 = \frac{2}{3} \times 4 + \frac{8}{9} \times 2^2 = \frac{56}{9}$

19.5 Since $N \sim \text{Poisson}(\lambda)$:

$$M_N(u) = \exp[\lambda(e^u - 1)]$$

So:

$$M_S(t) = M_N[\log M_X(t)] = \exp\left[\lambda\left(e^{\log M_X(t)} - 1\right)\right] = \exp\left[\lambda\left(M_X(t) - 1\right)\right]$$

Since $X \sim \text{Gamma}(\alpha, \beta)$:

$$M_X(t) = \left(1 - \frac{t}{\beta}\right)^{-\alpha}$$

So:

$$M_S(t) = \exp\left[\lambda\left(\left(1 - \frac{t}{\beta}\right)^{-\alpha} - 1\right)\right]$$

19.6 Option C is correct.

$S_1 + S_2$ has a compound Poisson distribution with Poisson parameter $(\lambda_1 + \lambda_2)$ and individual claim amount distribution $(\lambda_1 F_1(x) + \lambda_2 F_2(x))/(\lambda_1 + \lambda_2)$.

19.7 Let N_i denote the number of claims on policies of type i and let X_i denote the claim amount random variable for policies of type i , for $i = 1, 2$. Then:

$$M_{S_i}(t) = M_{N_i}(\ln M_{X_i}(t)) = \exp(\lambda_i(M_{X_i}(t) - 1))$$

By independence:

$$M_S(t) = E(e^{tS}) = E(e^{t(S_1 + S_2)}) = E(e^{tS_1} e^{tS_2}) = E(e^{tS_1}) E(e^{tS_2}) = M_{S_1}(t) M_{S_2}(t) \quad [1]$$

Hence:

$$\begin{aligned} M_S(t) &= \exp(\lambda_1(M_{X_1}(t) - 1)) \exp(\lambda_2(M_{X_2}(t) - 1)) \\ &= \exp(\lambda_1 M_{X_1}(t) + \lambda_2 M_{X_2}(t) - (\lambda_1 + \lambda_2)) \\ &= \exp(\lambda(M_W(t) - 1)) \end{aligned} \quad [1]$$

where:

$$\lambda = \lambda_1 + \lambda_2 \quad [1]$$

and:

$$M_W(t) = \frac{\lambda_1 M_{X_1}(t) + \lambda_2 M_{X_2}(t)}{\lambda_1 + \lambda_2} \quad [1]$$

Hence S is a compound Poisson random variable with Poisson parameter $\lambda = \lambda_1 + \lambda_2$.

19.8 (i) **MGF of S**

The MGF of S is:

$$M_S(t) = E(e^{tS}) = E[E(e^{tS} | N)]$$

using the standard result for conditional means from page 16 of the *Tables*.

Looking at the inner expression, we have:

$$E(e^{tS} | N = n) = E(e^{t(X_1 + \dots + X_n)}) = E(e^{tX_1}) \dots E(e^{tX_n}) \quad [1]$$

Now each of these terms is just the MGF of the random variable X .

So:

$$E(e^{tS} | N = n) = [M_X(t)]^n$$

and hence:

$$E(e^{tS} | N) = [M_X(t)]^N \quad [1]$$

So:

$$M_S(t) = E\left[\{M_X(t)\}^N\right] = E\left(e^{N \log M_X(t)}\right) = M_N[\log M_X(t)] \quad [1]$$

(ii) **Compound negative binomial distribution**

We first need the individual MGFs. Using results from the *Tables*, we have:

$$M_X(t) = \left(1 - \frac{t}{0.1}\right)^{-10} = (1 - 10t)^{-10} \quad [\frac{1}{2}]$$

$$\text{and: } M_N(t) = \left(\frac{0.9}{1 - 0.1e^t}\right)^2 \quad [\frac{1}{2}]$$

Combining these, using the result from part (i):

$$M_S(t) = M_N[\log M_X(t)] = \left(\frac{0.9}{1 - 0.1(1 - 10t)^{-10}}\right)^2 \quad [1]$$

(iii)(a) **Mean and variance of S**

We could differentiate this expression to find the mean and variance of S. However, it is much easier to use the standard compound distribution formulae:

$$E(S) = E(X)E(N)$$

$$\text{and: } \text{var}(S) = [E(X)]^2 \text{var}(N) + \text{var}(X)E(N)$$

Using the results from the *Tables* for the individual distributions:

$$\begin{aligned} E(X) &= \frac{\alpha}{\lambda} = \frac{10}{0.1} = 100 \\ \text{var}(X) &= \frac{\alpha}{\lambda^2} = \frac{10}{0.1^2} = 1,000 \end{aligned} \quad [1]$$

$$\begin{aligned} E(N) &= \frac{kq}{p} = \frac{2 \times 0.1}{0.9} = 0.22222 \\ \text{var}(N) &= \frac{kq}{p^2} = \frac{2 \times 0.1}{0.9^2} = 0.24691 \end{aligned} \quad [1]$$

Using the formulae above:

$$E(S) = E(X)E(N) = 22.222$$

$$\text{and: } \text{var}(S) = [E(X)]^2 \text{var}(N) + \text{var}(X)E(N) = 100^2 \times 0.24691 + 1,000 \times 0.22222 = 2,691.358 \quad [1]$$

(iii)(b) **Aggregate amount**

We now assume that S has an approximate normal distribution with this mean and variance. So, standardising in the usual way, we have:

$$P(S > k) = 0.001 \Rightarrow P\left(N(0,1) > \frac{k - E(S)}{\sqrt{\text{var}(S)}}\right) = 0.001$$

Using the percentage points of the standard normal distribution, we find that:

$$\frac{k - E(S)}{\sqrt{\text{var}(S)}} = 3.0902 \Rightarrow k = 22.222 + 3.0902\sqrt{2,691.358} = 182.54 \quad [1]$$

(iv)(a) **MGF of T**

The MGF of T is:

$$\begin{aligned} M_T(t) &= E(e^{tT}) = E[e^{t(S_1 + \dots + S_{100})}] = E(e^{tS_1}) \dots E(e^{tS_{100}}) = [M_S(t)]^{100} \\ &= \left(\frac{0.9}{1 - 0.1(1 - 10t)^{-10}} \right)^{200} \end{aligned} \quad [1]$$

The mean and variance of T are 100 times the corresponding results for S , ie:

$$E(T) = 100E(S) = 2,222$$

$$\text{and: } \text{var}(T) = 100 \text{var}(S) = 269,135.8 \quad [1]$$

(iv)(b) **Total aggregate amount**

So the corresponding figure for the aggregate amount exceeded with probability 0.001 is:

$$2,222 + 3.0902\sqrt{269,135.8} = 3,825.37 \quad [1]$$

(iv)(c) **Comment**

This is substantially less than one hundred times the corresponding answer to part (iii)(b). The Central Limit Theorem tells us that as the number of portfolios increases, bad experience in some of the portfolios will be offset by better experience in others, leading to a situation where the overall variation is relatively smaller. Pooling of similar risks reduces the overall variance. We can see this happening here. [1]